

MCQ:-

1. C
2. B
3. B
4. A
5. C
6. A
7. B
8. A
9. A
10. A
11. B
12. D

Q.2(i)

A.2(i)

Instrument:- A measuring tape should be used.

Reason:- A meter rule is only about 1m long. so measuring 12m with it would require repositioning it many times, introducing cumulative error. A measuring tape is flexible ribbon that can be pulled out to several metres in 1 continuous length, making

be measuring measurement faster and more accurate.

Precaution against parallax error:-
The eye must be positioned directly (perpendicularly) above/in line with the scale marking being read, not at an angle, otherwise the reading will appear shifted.

Q: 2(ii) (or)

A: 2(ii) (or)

$$\begin{aligned} \text{Pitch} &= 0.5 \text{ mm} \quad \text{circular scale reading: } 25^{\text{division}} \\ \text{L.C} &= \frac{\text{Pitch}}{\text{no of divisions}} = \frac{0.5}{25} \Rightarrow 0.02 \text{ mm} \end{aligned}$$

$$\text{C.S.R} : 15 \times 0.02 = 0.3 \text{ mm}$$

$$\text{Diameter: } \text{M.S.R} + \text{C.S.R} = 2 \text{ mm} + 0.3 \text{ mm} = 2.3 \text{ mm}$$

Q: 2(iii)

A: 2(iii)

$$u = 9 \text{ m/s}, \quad a = 2 \text{ m/s}^2, \quad t = 5 \text{ s}$$

(a)

$$v = u + at$$

$$= 9 + (2 \times 5)$$

$$= 9 + 10 = 19 \text{ m/s}$$

$$s = ut + \frac{1}{2}at^2$$

$$9 \times 5 + \frac{1}{2}(2)(5)^2$$

$$45 + \frac{1}{2}(2)(5)^2$$

$$45 + \frac{1}{2}(2)(5)^2$$

$$45 + 25 = 70 \text{ m.}$$

Q2(iv)(or)

A: 2(iv)(or)

Distance is the total length of the path travelled by a body, regardless of direction (a scalar quantity).
Displacement is the shortest straight line distance b/w the initial to final positions, along with direction (vector quantity).

a) Total distance: $5 + 12 + 5 = 22 \text{ km}$

b) Net displacement: 5 km north and 5 km south cancel out (she returns to her starting latitude) leaving only the 12 km eastward leg. Net displacement = 12 km , due east.

Q: 2(v)

A: 2(v)

Newton's second law:

The acceleration of an object is directly proportional to the magnitude of the net force, & in the same direction is inversely proportional to the mass of an object.

$$a = \frac{F}{m} = \frac{36}{6} = 6 \text{ m/s}^2$$

$$v = u + at = 0 + (6 \times 4) = 24 \text{ m/s}$$

Q:2(vi)

A:2(vi)

Momentum :

is the product of a body's mass and velocity ($p=mv$), a vector quantity. Impulse is the product of the average force and the time for which it acts ($\text{Impulse} = F \times t$) and it equals the change in momentum produced.

$$\textcircled{a} \Delta p = m(v-u) = 70 \times (0-15) = -1050 \text{ kg m/s}$$

$$\textcircled{b} F = \frac{\Delta p}{\Delta t} = \frac{-1050}{0.2} = -5250 \text{ N}$$

Q:2(vii)

A:2(vii)

Optics :

Branch of physics that studies the nature and properties of light, its interaction with matter, and instruments that use or detect it.

Example :

Eye glasses.

Classical mechanics:-

Branch of physics in which we study the motion of macroscopic object under the action of force.

Example:-

Calculating the trajectory of a thrown ball.

Quantum mechanics:-

Branch of physics in which we study the mathematical description of the motion and behaviour of microscopic objects under the action of force.

Q.2(viii)

A:2(viii)

Like parallel force:-

Forces that are parallel to each other and act in the same direction. Example:- 2 people pulling a stalled cart from behind with both hands, in the same direction.

Unlike parallel force:-

Forces that are parallel

to each other but act in different directions. Example:- turning bicycle handlebars where 1 hand may push forward while the others pull back (or applies a smaller force in the opposite senses).

Q: 2 (in)

A: 2 (in)

- Pascal principle :-

Pressure applied to an enclosed fluid is transmitted equally and undiminished to every point of the fluid and to the wall of its container.

Q: 2 (in)

A: 2 (in)

Pressure :-

Pressure is the force acting perpendicularly per unit area:

$$P = \frac{F}{A}$$

SI unit is ~~kg/m²~~ pascal (Pa)

which is N/m^2 .

$$F = 900 \text{ N} \quad \text{and} \quad A = 0.006 \text{ m}^2$$

$$P = \frac{F}{A} = \frac{900}{0.006} = 150,000 \text{ Pa} = 1.5 \times 10^5 \text{ Pa}$$

$$\text{Q: 2 (ni) (or)} \\ \text{A: 2 (ni) (or)}$$

Geothermal energy :-

Comes from the heat stored within the Earth's crust. Wells are drilled down to hot underground reservoirs; the heat converts water into steam, which is brought to the surface to drive turbines connected to generators, producing electricity.

$$\text{Q: 3 (i) (or)} \\ \text{A: 3 (i) (or)}$$

② The 3 equations of motion :-

$$\begin{aligned} * v &= u + at \\ * s &= ut + \frac{1}{2} at^2 \\ * v^2 &= u^2 + 2as \end{aligned} \quad \left\{ \begin{array}{l} u = \text{initial velocity} \\ v = \text{final velocity} \\ t = \text{time} \\ a = \text{acceleration} \end{array} \right.$$

$$(b) \quad v = u - gt = 0 = 18 - 10t$$

$$t = \frac{18}{10}$$

$$= 1.8 \text{ m/s.}$$

(c) maximum height :-

$$h = ut - \frac{1}{2}gt^2$$

$$= (18 \times 1.8) - \frac{1}{2}(10)(t)^2$$

$$16 \quad 32.4 - \frac{1}{2}(10)(1.8)^2$$

$$32.4 - (5)(1.8)^2$$

$$32.4 - (5)(3.24)$$

$$32.4 - 16.2 = 16.2 \text{ m}$$

Q: 3(ii) (or)

A: 3(iii) (or)

Q Before the explosion, the total momentum of the stationary body is zero. Since no external force acts during the explosion, momentum is conserved: The total momentum of the 2 fragments after explosion must also equal zero, so they fly apart in exactly opposite directions with momenta of equal magnitude.

Q Velocity of the 6 kg fragment.

$$m_1 v_1 + m_2 v_2 = 0$$

$$(4)(15) + (6)(v_2) = 0$$

$$60 + 6v_2 = 0$$

$$v_2 = \frac{-60}{6} \quad v_2 = -10 \text{ m/s}$$

$$KE = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$= \frac{1}{2} (4)(15)^2 + \frac{1}{2} (6)(10)^2$$

$$\frac{1}{2} (4)(225) + \frac{1}{2} (6)(100)$$

$$KE = 450 + 300$$

$$750 \text{ J}$$

Q: 3(iii)(or)

A: 3(iii)(or)

(a) Manometer :-

is a U-shaped tube, partially filled with a liquid used to measure the pressure of gas. One end of the tube is connected to the gas supply, and the other end is open to the atmosphere.

The gas pressure pushes the liquid down on the connected side up and up on the open side. At equilibrium, the difference in pressure b/w the gas and the atmosphere equals the pressure exerted by the difference in liquid column heights (h) b/w 2 limbs:

Numerical :-

$$\rho = 950 \text{ kg/m}^3 \quad h = 32 \text{ cm} = 0.32 \text{ m}$$

$$\text{atmospheric } (P) = 1.0 \times 10^5 \text{ Pa}$$

$$g = 10$$

$$\textcircled{1} P_{\text{diff}} = \rho gh = 950 \times 10 \times 0.32$$

$$3040 \text{ Pa}$$

$$\textcircled{2} \text{ Absolute gas pressure} = \text{Atmospheric pressure} + P_{\text{diff}}$$

$$= 100,000 + 3040$$

$$103,040 \text{ Pa}$$

Q: 3(iv)

A: 3(iv)

a) Law of conservation of energy :-
Energy can neither be created nor destroyed
; PE can only be converted from 1 form to
another, and the total energy of an isolated
system remain constant.

b) Numerical :-

$$\textcircled{i} \text{ PE at the top} - E_p = mgh = 6 \times 10 \times 20 = 1200$$

$$\textcircled{ii} K_E = E_p = 1200 \text{ J}$$

$$\frac{1}{2} mv^2 = 1200$$

$$v^2 = \frac{2 \times 1200}{6} = 400 \Rightarrow \sqrt{v^2} = \sqrt{400}$$

$$v = 20 \text{ m/s}$$